

Practical Exercise :SQL Null Functions

Question 1

Run all (1000) Just now (8s) workspace: default New SQL editor: ON

```
-----1.Find all records where Size is missing and the purchase_amount is greater than 50. Expected Columns: Customer ID, Size, purchase_amount, Item Purchased
-----
select `Customer ID`,
       Size,
       `purchase_amount`,
       `Item Purchased`
from `workspace`.`default`.`shoping_trends`
where Size is null
and
       `purchase_amount` > 50;
```

Add parameter

	Customer ID	Size	purchase_amount	Item Purchased
1	11	null	74	Handbag
2	15	null	54	Jeans
3	22	null	88	Shirt
4	32	null	54	Blouse

22 rows | 7.87s runtime Refreshed now

Question 2

Run all (1000) Just now (7s) workspace: default New SQL editor: ON

```
-----2.List the total number of purchases grouped by Season, treating NULL values as 'Unknown Season'. Expected Columns: Season, Total Purchases
-----
select IFNULL(`Season`, 'Unknown Season') as Season,
       count(*) as Total_Purchases
from `workspace`.`default`.`shoping_trends`
GROUP by Season;
```

Add parameter

	Season	Total_Purchases
1	Winter	80
2	Spring	73
3	Unknown Seas...	27
4	Summer	65
5	Fall	55

5 rows | 6.85s runtime Refreshed now

Question 3

Run selected (1000) Just now (1s) workspace default New SQL editor: ON

```

20
21 ---3.Count how many customers used each Payment Method, treating NULLs as 'Not Provided'. Expected Columns: Payment Method, Customer Count
22
23 select IFNULL(`Payment Method`, 'Not Provided') as Payment_Method,
24        count(distinct `Customer ID`) AS Customer_Count
25 from `workspace`.`default`.`shopping_trends`
26 GROUP BY Payment_Method;
27

```

Add parameter

	Payment_Method	Customer_Count
1	PayPal	51
2	Debit Card	42
3	Not Provided	30
4	Bank Transfer	38
5	Venmo	53
6	Credit Card	44

7 rows | 0.91s runtime Refreshed now

[See performance \(1\)](#) [Optimize](#)

Question 4

New Query 2025-11-17 11:32am

Run all (1000) Just now (<1s) workspace default New SQL editor: ON

```

28
29 ---4.Show customers where Promo Code Used is NULL and Review Rating is below 3.0. Expected Columns: Customer ID, Promo Code Used, Review
Rating, Item Purchased
30
31 SELECT `Customer ID`,
32        `Promo Code Used`,
33        `Review Rating`,
34        `Item Purchased`
35 from `workspace`.`default`.`shopping_trends`
36 WHERE `Promo Code Used` IS NULL
37 AND `Review Rating` < 3.0;
38

```

Add parameter

	Customer ID	Promo Code Used	Review Rating	Item Purchased
1	21	null	2.5	Jeans
2	38	null	2.6	Jeans
3	61	null	2.5	Jeans
4	80	null	2.6	Sneakers

8 rows | 0.20s runtime Refreshed now

[See performance \(1\)](#) [Optimize](#)

Question 5

```

37
38 -----
39 --5.Group customers by Shipping Type, and return the average purchase_amount, treating missing values as 0. Expected Columns: Shipping
40 Type, Average purchase_amount
41 -----
42 select `Shipping Type`,
43        avg(IFNULL(`purchase_amount`,0)) AS Average_purchase_amount
44 from `workspace`.`default`.`shopping_trends`
45 GROUP BY `Shipping Type`;

```

Add parameter

	Shipping Type	Average_purchase_amount
1	Standard	47.666666666666664
2	Store Pickup	55.333333333333336
3	null	52.7037037037037
4	Express	53.45454545454545
5	2-Day Shipping	51.55769230769231

7 rows | 0.17s runtime

Refreshed 2 minutes ago

Question 6

New Query 2025-11-17 11:32am ×

Run all (1000) ✓ Just now (<1s) workspace default New SQL editor: ON ☆

```

45 -----
46 --6.Display the number of purchases per Location only for those with more than 5 purchases and no NULL Payment Method. Expected Columns:
47 Location, Total Purchases
48 -----
49 > select Location, ...
50 from `workspace`.`default`.`shopping_trends`
51 WHERE `Payment Method` IS NOT null
52 GROUP BY Location
53 HAVING `total_purchases` > 5;
54
55

```

Add parameter

	Location	Total_Purchases
1	Maine	41
2	Rhode Island	29
3	null	24
4	Oregon	30
5	Florida	32

9 rows | 0.17s runtime

Refreshed now

Question 7

```
--7.Create a column Spender Category that classifies customers using CASE: 'High' if amount > 80, 'Medium' if BETWEEN 50 AND 80, 'Low' otherwise. Replace NULLs in purchase_amount with 0. Expected Columns: Customer ID, purchase_amount, Spender Category
```

```
select `Customer ID`,
       `purchase_amount`,
       CASE WHEN `purchase_amount` > 80 THEN 'High'
            WHEN `purchase_amount` BETWEEN 50 AND 80 THEN 'Medium'
            ELSE 'Low'
       END as Spender_Category
from `workspace`.`default`.`shopping_trends`;
```

parameter

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Q 🔍 📄 📊 🔄 Preview 📄 ^ ×

1.3 Customer ID	1.2 purchase_amount	1.0 Spender_Category
4	45	Low
5	80	Medium
6	82	High
7	50	Medium

300 rows | 0.81s runtime

Refreshed 2 minutes ago

Question 8

```
--8.Find customers who have no Previous Purchases value but whose Color is not NULL. Expected Columns: Customer ID, Color, Previous Purchases
```

```
SELECT `customer ID`,
       `Color`,
       `Previous Purchases`
FROM `workspace`.`default`.`shopping_trends`
WHERE `Previous Purchases` IS NULL
AND `Color` IS NOT NULL;
```

Add parameter

Table ▾ +

Q 🔍 📄 📊 🔄 Preview 📄 ^ ×

1.3 customer ID	1.0 Color	1.2 Previous Purchases
1	8	Green
2	21	Yellow
3	25	White
4	37	Maroon

36 rows | 0.49s runtime

Refreshed now

Question 9

```

74  -----9.Group records by Frequency of Purchases and show the total amount spent per group, treating NULL frequencies as 'Unknown'. Expected
    Columns: Frequency of Purchases, Total purchase_amount
75  -----
76
77  SELECT IFNULL(`Frequency of Purchases`, 'Unknown') as Frequency_of_Purchases,
78  -----SUM(`purchase_amount`) as Total_purchase_amount
79  FROM `workspace`.`default`.`shopping_trends`
80  GROUP BY Frequency_of_Purchases;
81
82

```

Add parameter

	Frequency_of_Purchases	Total_purchase_amount
1	Every 3 Months	1749
2	Weekly	2184
3	Bi-Weekly	2099
4	Monthly	1780

8 rows | 0.06s runtime

Refreshed now

[See performance \(1\)](#) [Optimize](#)

Question 10

Run all (1000) ✓ Just now (<1s) workspace default New SQL editor: ON ☆

```

82  -----10.Display a list of all Category values with the number of times each was purchased, excluding rows where Category is NULL. Expected
    Columns: Category, Total Purchases
83  -----
84  SELECT `Category`,
85  count(DISTINCT `Customer ID`) AS Total_Purchases
86  FROM `workspace`.`default`.`shopping_trends`
87  WHERE `Category` IS NOT NULL
88  GROUP BY `Category`;

```

Add parameter

	Category	Total_Purchases
1	Footwear	70
2	Outerwear	60
3	Clothing	59
4	Accessories	78

4 rows | 0.08s runtime

Refreshed now

Question 11

New Query 2025-11-17 11:32am x +

Run all (1000) ✓ 2 minutes ago (14s) workspace default New SQL editor: ON ☆

```
91 -----11.Return the top 5 Locations with the highest total purchase_amount, replacing NULLs in amount with 0. Expected Columns: Location,
92 Total_purchase_amount
93 SELECT `Location`,
94       SUM(IFNULL(`purchase_amount`,0)) AS Total_purchase_amount
95 FROM `workspace`.`default`.`shopping_trends`
96 GROUP BY `Location`
97 ORDER BY Total_purchase_amount DESC
98 LIMIT 5;
```

Add parameter

Table +

	Location	Total_purchase_amount
1	Maine	2294
2	Florida	1980
3	Massachusetts	1899
4	Rhode Island	1876

5 rows | 14.32s runtime Refreshed 1 minute ago

> See performance (1) Optimize

Question 12

New Query 2025-11-17 11:32am x +

Run selected (1000) ✓ Just now (2s) workspace default New SQL editor: ON ☆

```
97 ORDER BY Total_purchase_amount DESC
98 LIMIT 5;
99
100 -----12.Group customers by Gender and Size, and count how many entries have a NULL Color. Expected Columns: Gender, Size, Null Color Count
101 SELECT `Gender`,
102        `Size`,
103        count(DISTINCT `Customer_ID`) AS Null_Color_Count
104 FROM `workspace`.`default`.`shopping_trends`
105 WHERE `Color` IS NULL
106 GROUP BY `Gender`,`Size`;
```

Add parameter

Table +

	Gender	Size	Null_Color_Count
1	Male	M	7
2	Male	small	6
3	Male	S	5

5 rows | 1.53s runtime Refreshed now

> See performance (1) Optimize

Question 13

New Query 2025-11-17 11:32am × +

Run selected (1000) ✓ 1 minute ago (<1s) workspace: default New SQL editor: ON ☆ Edit

```
--13. Identify all Item Purchased where more than 3 purchases had NULL Shipping Type. Expected Columns: Item Purchased, NULL Shipping Type Count
-----
110 SELECT `Item Purchased`,
111       count(DISTINCT `Customer ID`) AS Null_Shipping_Type_Count
112 FROM `workspace`.`default`.`shopping_trends`
113 WHERE `Shipping Type` IS NULL
114 GROUP BY `Item Purchased`
115 HAVING Null_Shipping_Type_Count > 3;
116
```

Add parameter

	Item Purchased	Null_Shipping_Type_Count
1	null	4
2	Shoes	4
3	Shirt	5

3 rows | 0.08s runtime Refreshed 1 minute ago

> [See performance \(1\)](#) [Optimize](#)

Question 14

New Query 2025-11-17 11:32am × +

Run all (1000) ✓ Just now (1s) workspace: default New SQL editor: ON ☆ Edit

```
---14. Show a count of how many customers per Payment Method have NULL Review Rating. Expected Columns: Payment Method, Missing Review Rating Count
-----
120 SELECT `Payment Method`,
121       count(DISTINCT `Customer ID`) AS Missing_Review_Rating
122 FROM `workspace`.`default`.`shopping_trends`
123 WHERE `Review Rating` IS NULL
124 GROUP BY `Payment Method`;
125
```

Add parameter

	Payment Method	Missing_Review_Rating
1	Credit Card	8
2	Cash	4
3	Bank Transfer	4
4	Debit Card	7

7 rows | 0.73s runtime Refreshed now

> [See performance \(1\)](#) [Optimize](#)

Question 15

New Query 2025-11-17 11:32am x +

Run selected (1000) Just now (<1s) workspace default New SQL editor: ON

```

125 --15.Group by Category and return the average Review Rating, replacing NULLs with 0, and filter only where average is greater than 3.5.
126 Expected Columns: Category, Average Review Rating
127
128 SELECT `Category`,
129        .....AVG(IFNULL(`Review_Rating`,0)) AS Average_Review_Rating
130 FROM `workspace`.`default`.`shopping_trends`
131 GROUP BY `Category`
132 HAVING Average_Review_Rating > 3.5;
133

```

Add parameter

Table	1.2 Average_Review_Rating
No rows returned	

0 rows | 0.06s runtime Refreshed now

[See performance \(1\)](#) [Optimize](#)

Question 16

New Query 2025-11-17 11:32am x +

Run selected (1000) Just now (<1s) workspace default New SQL editor: ON

```

135 --16.List all Colors that are missing (NULL) in at least 2 rows and the average Age of customers for those rows. Expected Columns:
136 Color, Average Age
137
138 SELECT `Color`,
139        .....AVG(Age) AS `Average Age`
140 FROM `workspace`.`default`.`shopping_trends`
141 WHERE `Color` IS NULL
142 GROUP BY `Color`
143 LIMIT 2;

```

Add parameter

Table	1.2 Average Age
1	47.84615384615385

1 row | 0.06s runtime Refreshed now

[See performance \(1\)](#) [Optimize](#)

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Question 17

Run all (1000) ✓ 1 minute ago (<1s) workspace: default New SQL editor: ON

```

143 --17.Use CASE to create a column Delivery Speed: 'Fast' if Shipping Type is 'Express' or 'Next Day Air', 'Slow' if 'Standard', 'Other'
144 for all else including NULL. Then count how many customers fall into each category. Expected Columns: Delivery Speed, Customer Count
145
146 SELECT
147     CASE
148         WHEN `Shipping Type` = 'Express' OR `Shipping Type` = 'Next Day Air' THEN 'Fast'
149         WHEN `Shipping Type` = 'Standard' THEN 'Slow'
150         ELSE 'Other'
151     END AS `Delivery Speed`,
152     COUNT(DISTINCT `Customer ID`) AS `Customer Count`

```

Add parameter

	Delivery Speed	Customer Count
1	Slow	45
2	Other	166
3	Fast	89

3 rows | 0.06s runtime Refreshed 1 minute ago

[See performance \(1\)](#) [Optimize](#)

Question 18

New Query 2025-11-17 11:32am x

Run selected (1000) ✓ Just now (<1s) workspace: default New SQL editor: ON

```

152 GROUP BY
153 CASE
154     WHEN `Shipping Type` = 'Express' OR `Shipping Type` = 'Next Day Air' THEN 'Fast'
155     WHEN `Shipping Type` = 'Standard' THEN 'Slow'
156     ELSE 'Other'
157 END;
158
159 --18.Find customers whose purchase_amount is NULL and whose Promo Code Used is 'Yes'. Expected Columns: Customer ID, purchase_amount,
160 Promo Code Used

```

Add parameter

	Customer ID	purchase_amount	Promo Code Used
1	13	null	Yes
2	30	null	Yes
3	78	null	Yes
4	95	null	Yes
5	124	null	Yes

20 rows | 0.09s runtime Refreshed now

[See performance \(1\)](#) [Optimize](#)

Question 19

New Query 2025-11-17 11:32am x +

Run all (1000) Just now (<1s) workspace default New SQL editor: ON Edit

```

166 --19.Group by Location and show the maximum Previous Purchases, replacing NULLs with 0, only where the average rating is above 4.0.
167 Expected Columns: Location, Max Previous Purchases, Average Review Rating
168 -----
169 SELECT `Location`,
170        MAX(IFNULL(`Previous Purchases`,0)) AS `Max Previous Purchase`,
171        AVG(`Review Rating`) AS `Average Review Rating`
172 FROM `workspace`.`default`.`shoping_trends`
173 GROUP BY `Location`
174 HAVING `Average Review Rating` > 4.0;

```

Add parameter

Table +

Location	1.2 Max Previous Purchase	1.2 Average Review Rating
No rows returned		

0 rows | 0.05s runtime Refreshed now

> See performance (1) Optimize

Question 20

New Query 2025-11-17 11:32am x +

Run all (1000) 1 minute ago (1s) workspace default New SQL editor: ON Edit

```

HAVING `Average Review Rating` > 4.0;
-----
--20.Show customers who have a NULL Shipping Type but made a purchase in the range of 30 to 70 USD. Expected Columns: Customer ID, Shipping
Type, purchase_amount, Item Purchased
-----
SELECT `Customer ID`,
       `Shipping Type`,
       `purchase_amount`,
       `Item Purchased`
FROM `workspace`.`default`.`shoping_trends`
WHERE `Shipping Type` IS NULL
AND `purchase_Amount` BETWEEN 30 AND 70;

```

Add parameter

Table +

	Customer ID	Shipping Type	1.2 purchase_amount	Item Purchased
1	15	null	54	Jeans
2	105	null	43	Shirt
3	141	null	37	Shorts
4	196	null	66	Coat

7 rows | 0.86s runtime Refreshed 1 minute ago

> See performance (1) Optimize